Sahaj Bhadja

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Summary

Software engineer with 2+ years building CI/CD pipelines, cross-platform test automation, cloud services, and reliable release workflows using Python, Bash, Jenkins, GitHub Actions, Azure, AWS, and Docker.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Built CI/CD-backed FastAPI integration workflows processing 5,000+ monthly jobs with validation, reconciliation, failure isolation, and monitored releases, achieving 99%+ reliability across recurring production workloads at scale.
- Architected Azure SQL and Blob Storage services for 500+ users, tuning indexed queries and access patterns to reduce dashboard latency from 1.2 seconds to 350 milliseconds in production.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Implemented GitHub Actions deployment workflows for AWS-hosted Flask services handling 10,000+ daily grading requests, reducing feedback turnaround by 60% while maintaining stable API performance during peaks.
- Designed concurrent batch automation for 1,000+ uploaded files using multithreading, secure ingestion, AWS RDS, and fault-tolerant execution, improving throughput and isolating failures across grading workflows at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Automated nightly Jenkins pipelines for builds, VM installation, and 20+ PyTest and Mockito suites across Windows, Linux, and macOS, generating coverage reports and actionable failure logs after nightly execution.
- Troubleshot REST API and AWS Glue ETL workflows using S3 and SQL, improving response times by 30%, pipeline efficiency by 68%, and release maintainability across distributed teams.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a MERN application with REST APIs, WebSockets, and eight modular services supporting 2M+ synchronized records while translating requirements into responsive customer and inventory workflows across production environments.
- Automated Docker-based builds, tests, and deployments through Jenkins CI/CD, reducing release time by 50% while improving integration consistency, defect resolution, and production delivery reliability for engineering teams.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built Flask APIs for a booking application supporting search, scheduling, payments, and status updates across 50,000+ records with synchronized Firebase-backed web and mobile workflows under concurrent workloads.
- Implemented webhook-based payment synchronization and operational troubleshooting, reducing manual payment handling by 70% while improving transaction visibility, error recovery, and responsiveness across distributed application components during production operations.

Skills

Automation & DevOps: Jenkins, GitHub Actions, CI/CD, build automation, release management, test automation, VM-based testing, Docker, Kubernetes, Git, Maven, Yarn, monitoring, debugging, failure isolation; **Cloud & Platforms:** Azure, Azure SQL, AWS, Amazon S3, AWS Glue, AWS RDS, GCP, Windows, Linux, macOS; **Programming:** Python, Bash, Java, C++, C, C#, PowerShell, JavaScript, TypeScript, SQL; **Engineering:** PyTest, Mockito, unit testing, integration testing, REST APIs, microservices, Agile/Scrum, object-oriented design, distributed workflows, performance optimization

Projects

Linguistix | *Python, AWS ECS/Fargate, Docker, CodePipeline, CI/CD*
- Developed and deployed a containerized data-processing API on AWS with automated CodePipeline releases, validation checks, and Docker-based builds, achieving 87% accuracy and production-grade reliability across repeatable environments.

Chat2SQL Code | *Java, TypeScript, Docker, Maven, Node.js, Yarn*
- Built a cross-platform database workspace supporting 10+ systems, using Docker, Maven, Node.js, and Yarn to standardize development and deployment workflows across Windows, macOS, and Linux for developers.

CPU-Scheduling Simulator Live | *C++, React, JavaScript, EJ-2*
- Implemented six scheduling algorithms with dynamic metrics and Gantt visualization across 1,000+ process combinations, reducing debugging time by 20% through deterministic simulation and systematic edge-case testing.